



CSE 432 Robotics

Rigid Motions and Homogeneous Transformations 1

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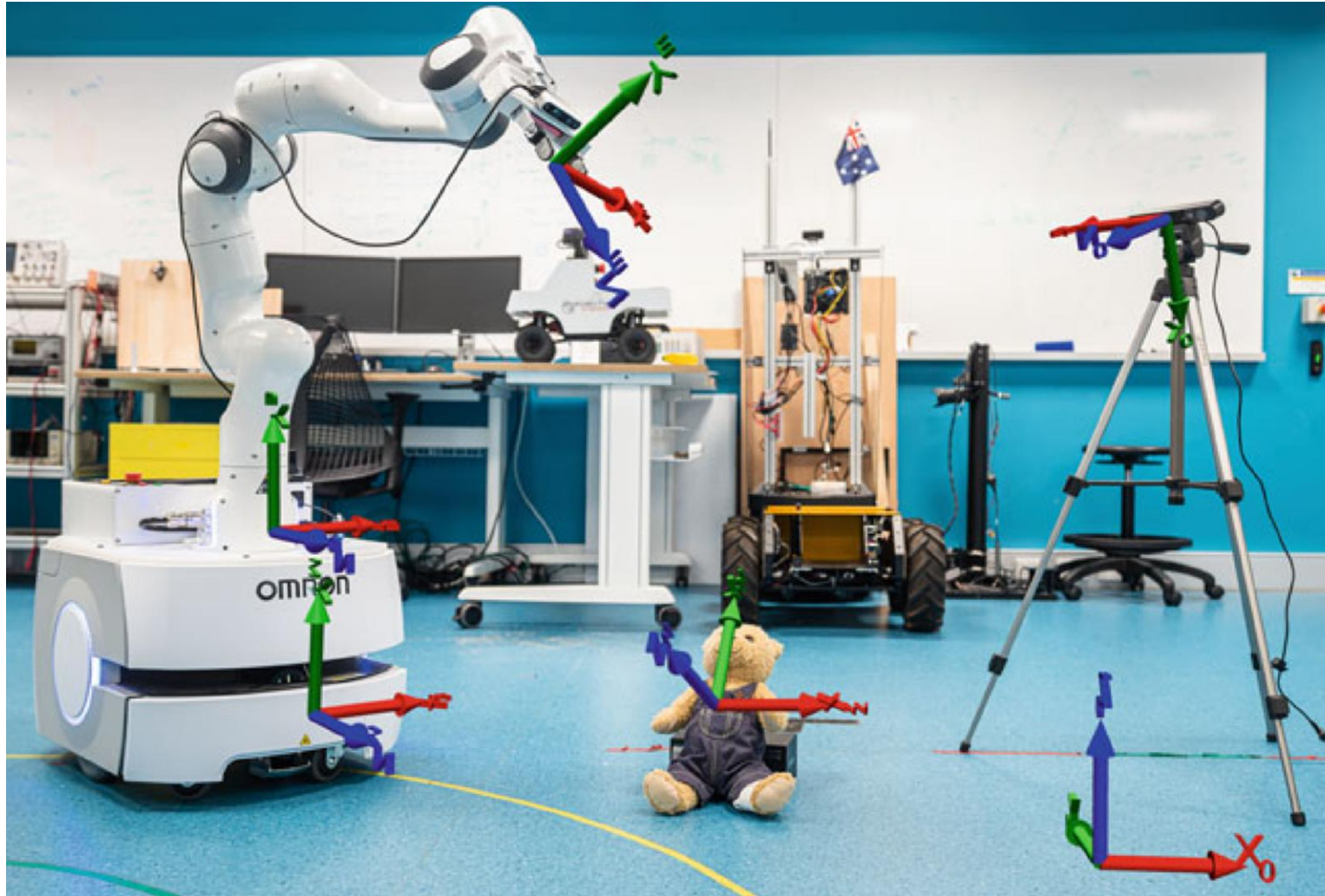
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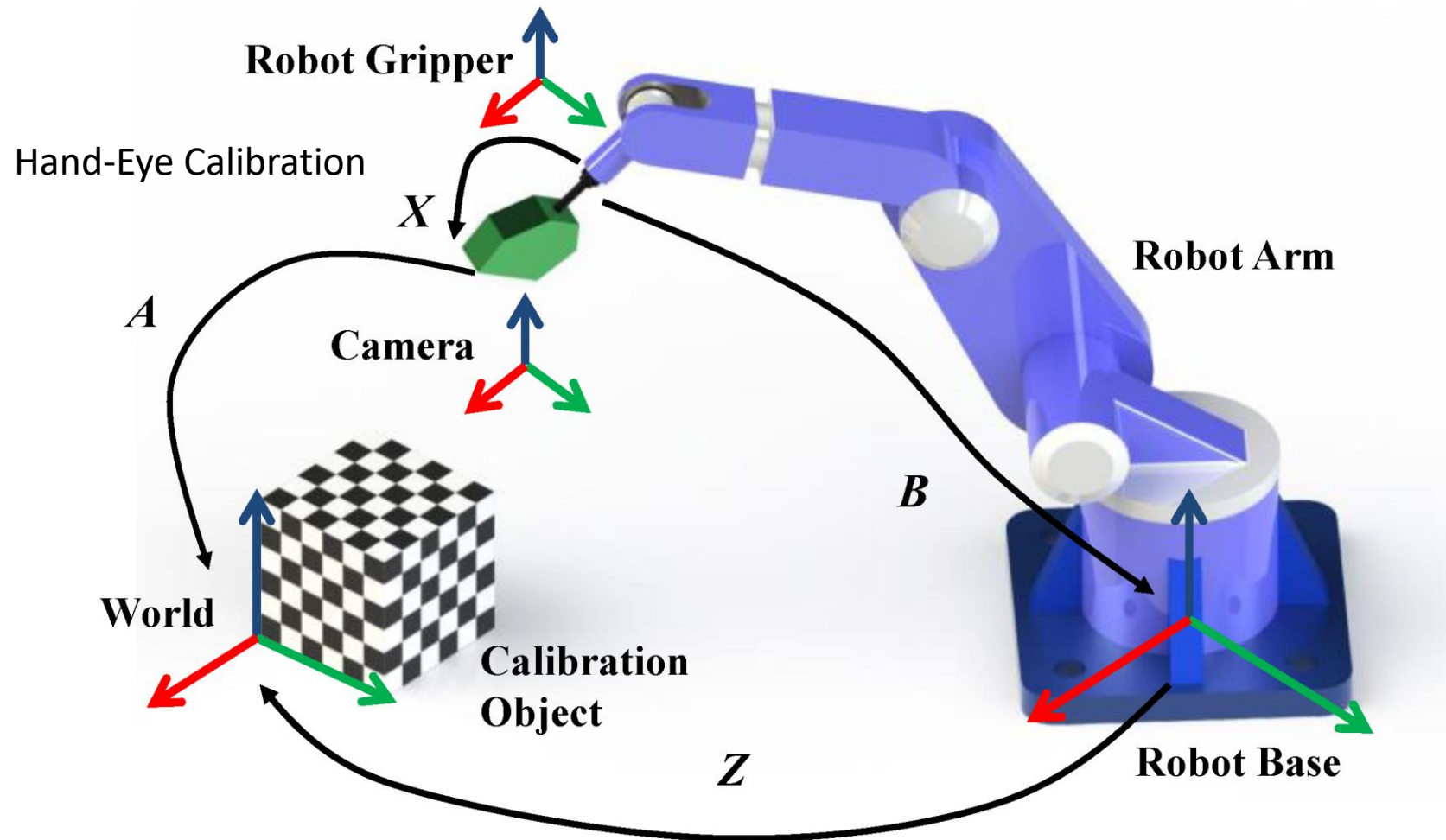


Introduction





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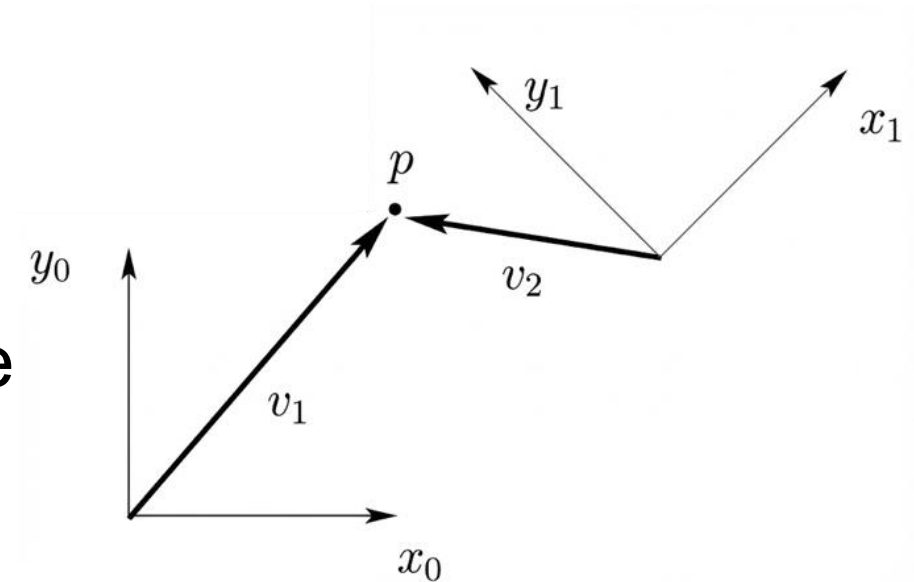




REPRESENTING POSITIONS

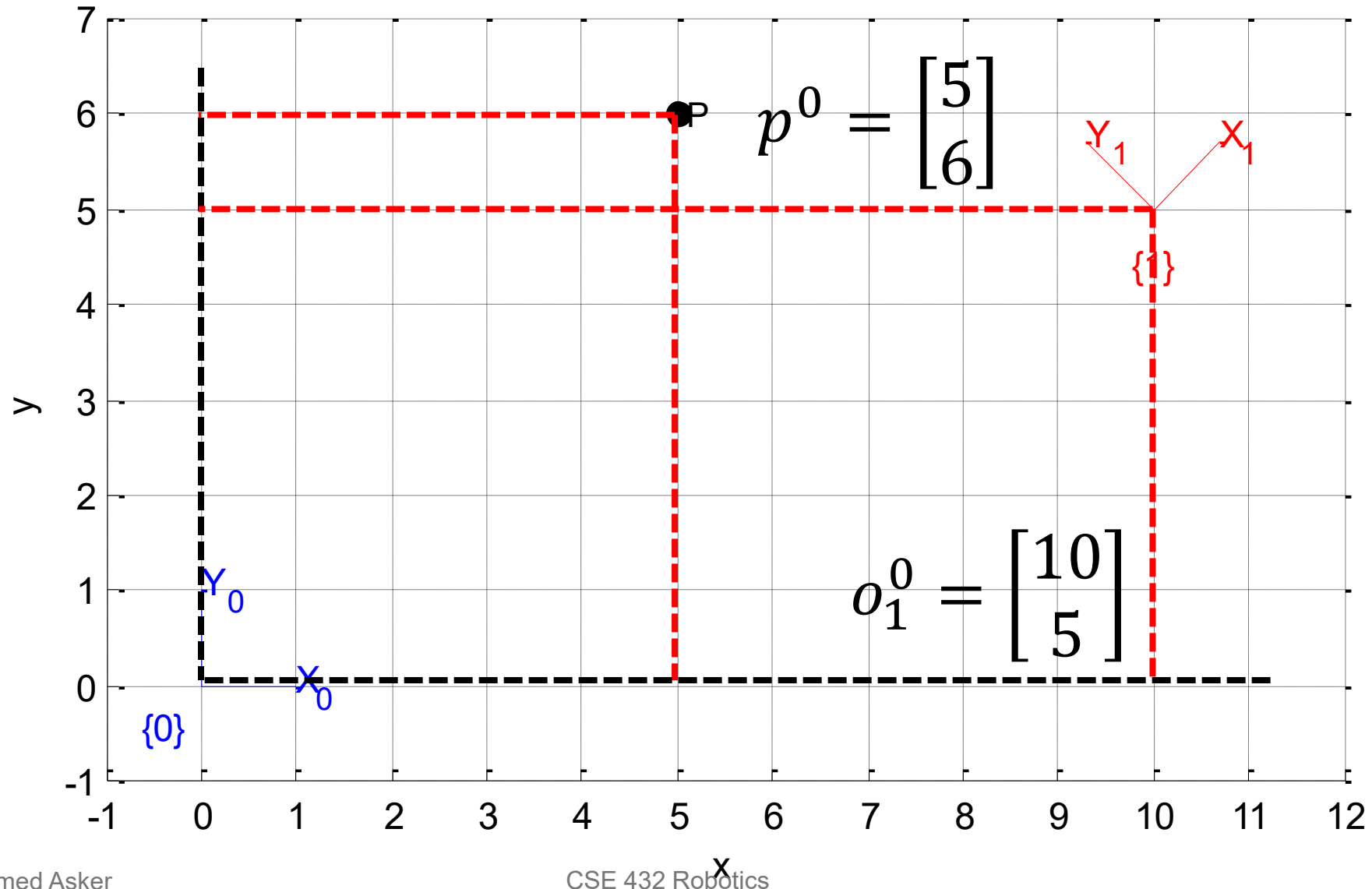
- A point p can be expressed in either frame $o_0x_0y_0$ or frame $o_1x_1y_1$.
- **For example:**
 - In frame $o_0x_0y_0$: $p^0 = (5, 6)$
 - In frame $o_1x_1y_1$: $p^1 = (-2.8, 4.2)$
- Use a superscript to identify the reference
- Thus, we write:

$$p^0 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, \quad p^1 = \begin{bmatrix} -2.8 \\ 4.2 \end{bmatrix}$$



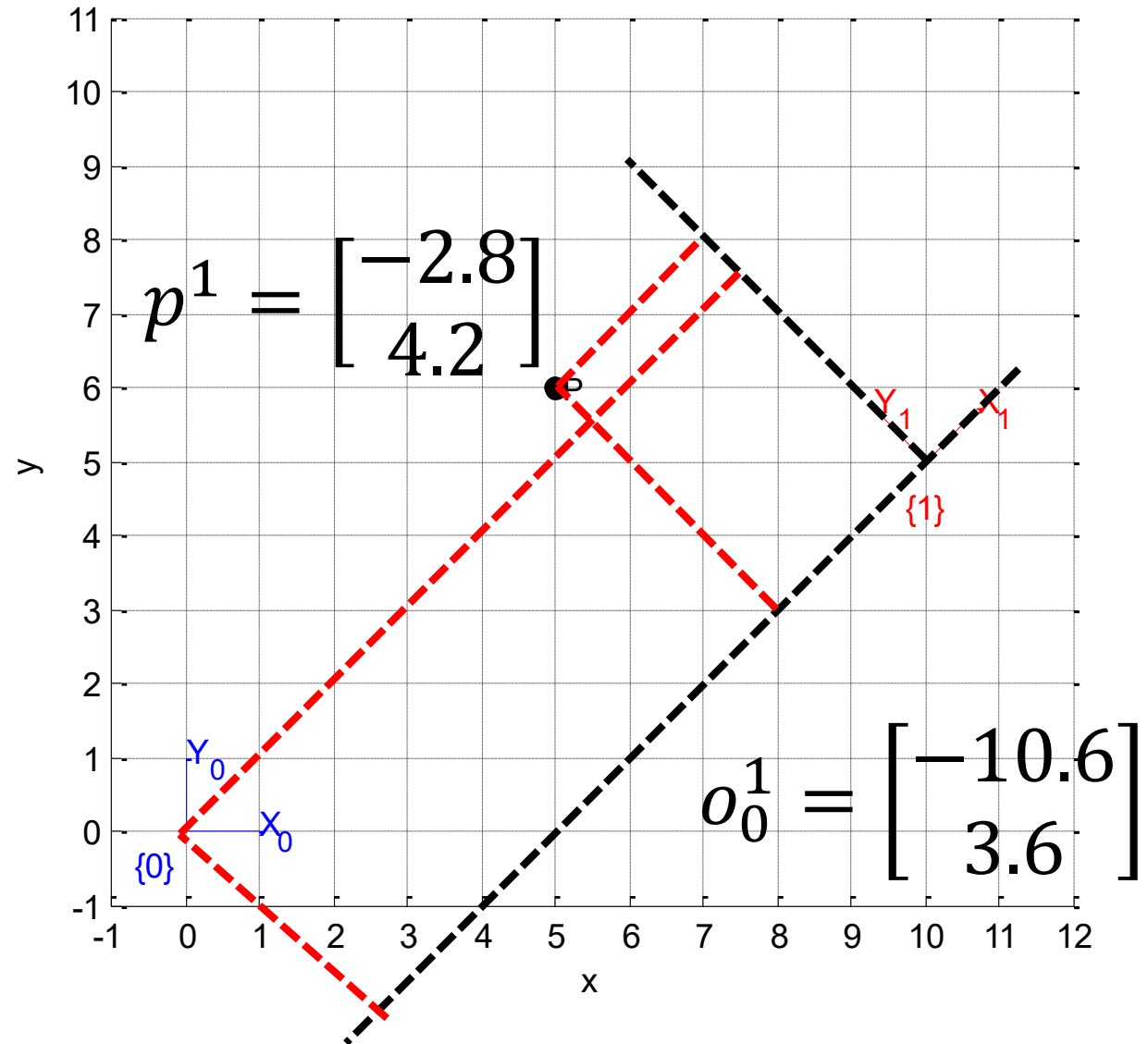


REPRESENTING POSITIONS





REPRESENTING POSITIONS





Coordinate Convention

- **All vectors must be expressed in the same reference frame.**
- For free vectors (e.g., force, velocity, etc.), it is sufficient that they be expressed in **parallel coordinate frames**.
- Only the **magnitude and direction** of a free vector matter.
- Frames with parallel (but not necessarily coincident) axis are adequate for defining free vectors.
- Thus, the expression of the form $v_1^1 + v_2^2$.
 - frames $o_0x_0y_0$ and $o_1x_1y_1$ are not parallel.



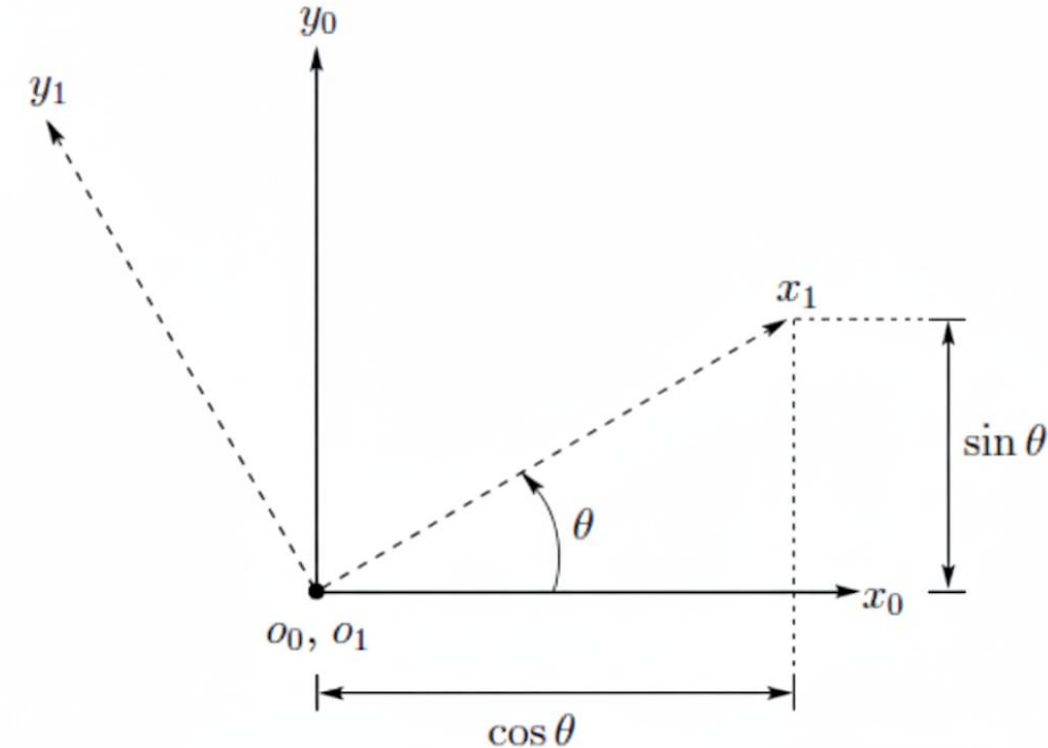
Rotation in the plane

- Orientation of frame $o_1x_1y_1$ with respect to coordinate frame $o_0x_0y_0$, can be expressed as follows:

$$x_1^0 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, y_1^0 = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

- In 2D, the rotation matrix:

$$R_1^0 = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$





Rotation in the plane

- A scalable 3D approach is to construct the rotation matrix by projecting the axes of frame $o_1x_1y_1$ onto the axes of frame $o_0x_0y_0$.

$$R_0^1 = \begin{bmatrix} x_0x_1 & y_0x_1 \\ x_0y_1 & y_0y_1 \end{bmatrix}$$

- Since dot product commutates

$$R_1^0 = \begin{bmatrix} x_1x_0 & y_1x_0 \\ x_1y_0 & y_1y_0 \end{bmatrix}$$



Rotation in the plane

Properties of Rotation Matrix

- $R^{-1} = R^T$
 - The columns (and therefore the rows) of R are mutually orthogonal
 - Each column (and therefore each row) of R is a unit vector
 - $\det R = 1$
-
- These properties apply to rotation matrices in three or more dimensions.



Rotations in three dimensions

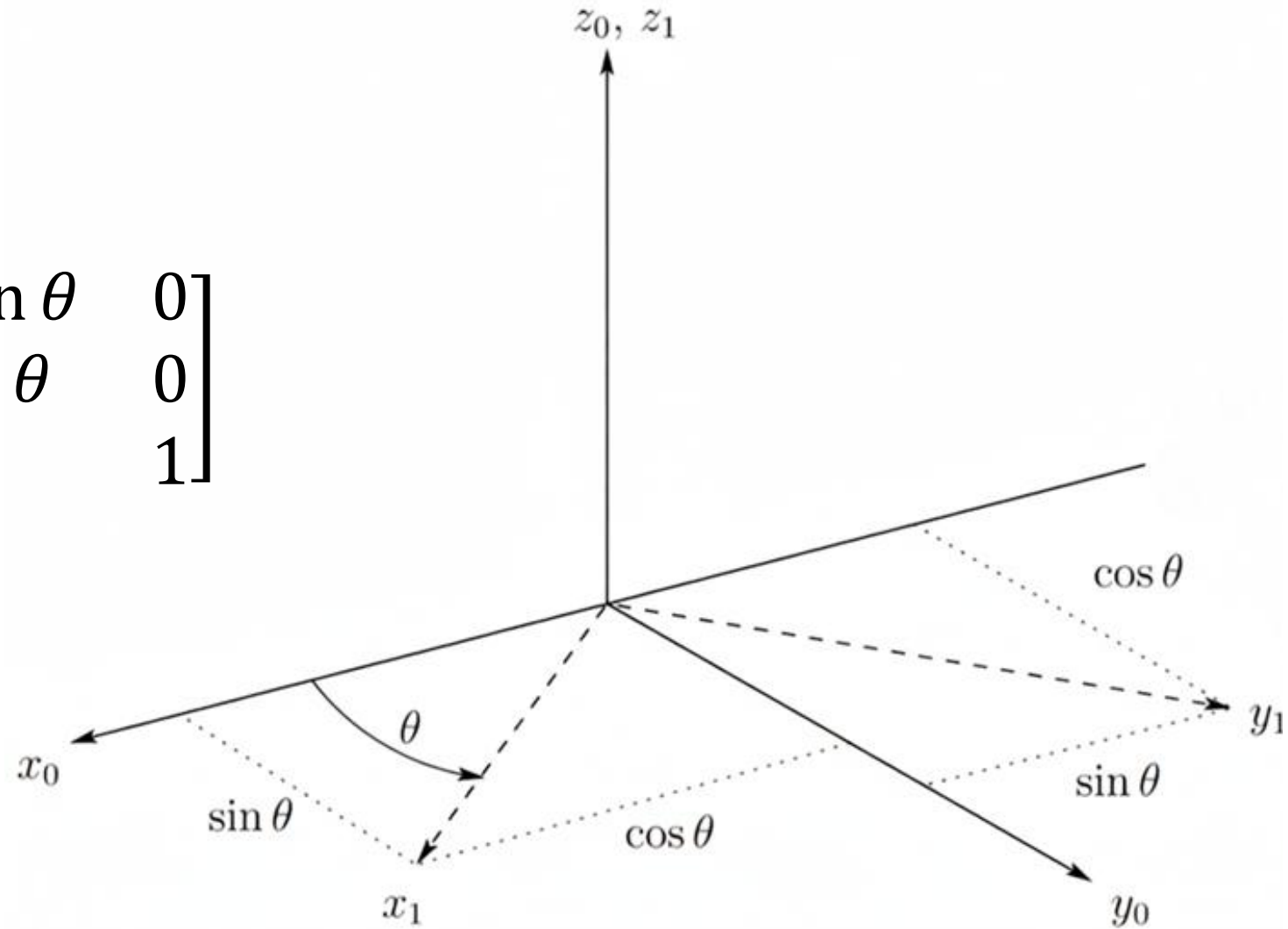
- The projection technique described above scales nicely to the 3D case.
- Each axis of the frame $o_1x_1y_1z_1$ is projected onto coordinate frame $o_0x_0y_0z_0$.
- The resulting rotation matrix is given by

$$R_1^0 = \begin{bmatrix} x_1x_0 & y_1x_0 & z_1x_0 \\ x_1y_0 & y_1y_0 & z_1y_0 \\ x_1z_0 & y_1z_0 & z_1z_0 \end{bmatrix}$$



Example 1

$$R = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$





The Basic Rotation Matrices

- *Rotation about z- axis by angle θ*

$$R_{z,\theta} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- *Rotation about x- axis by angle θ*

$$R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

- *Rotation about y- axis by angle θ*

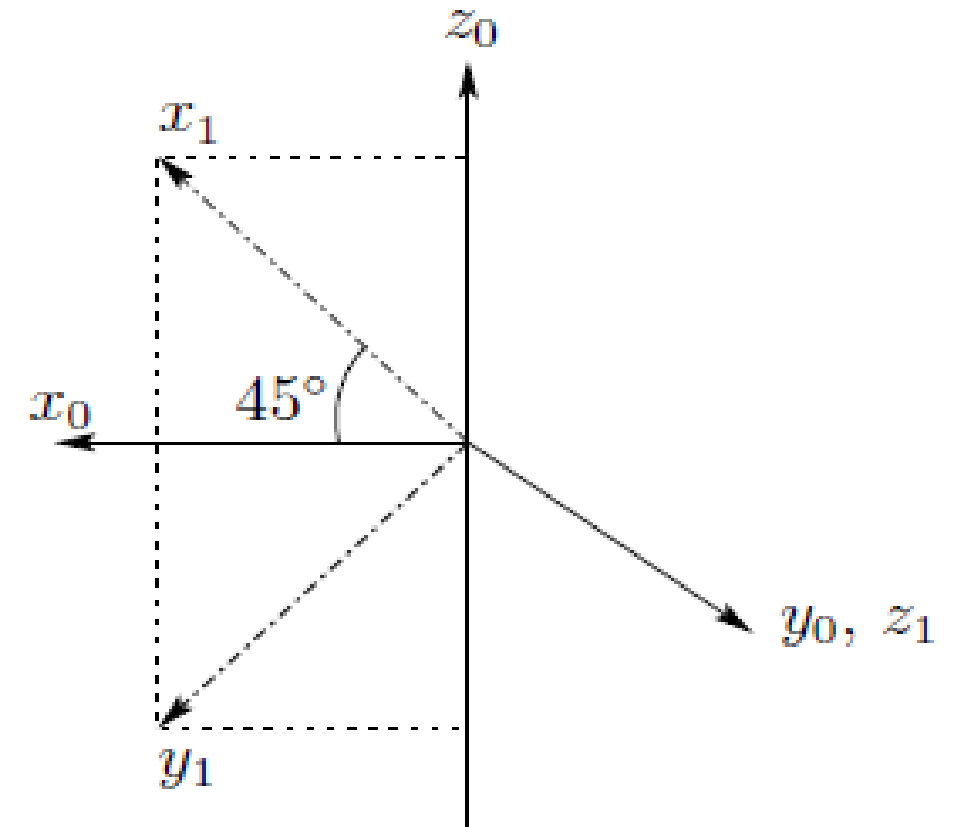
$$R_{y,\theta} = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$



Example 2

- Consider the frames $o_0x_0y_0z_0$ and $o_1x_1y_1z_1$.
- Find the rotation matrix R_1^0 .

$$R_1^0 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix}$$



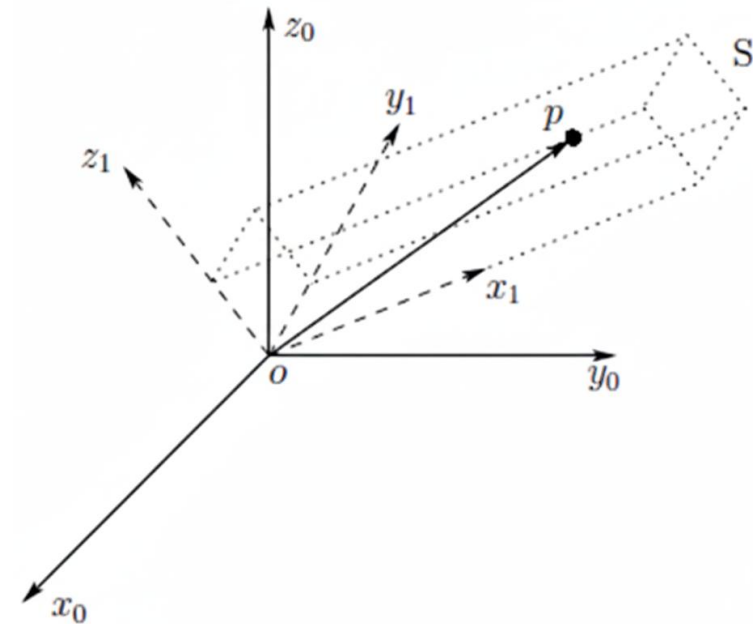


ROTATIONAL TRANSFORMATIONS

- A rigid object S to which a coordinate frame $o_1x_1y_1z_1$ is attached.
- Given the coordinates p^1 of point p .
- The coordinates $p_1 = (u, v, w)$ satisfy the equation:
$$p = ux_1 + vy_1 + wz_1$$

- p^0 can be obtained by projecting point p onto frame $o_0x_0y_0z_0$, giving

$$p^0 = \begin{bmatrix} x_1x_0 & y_1x_0 & z_1x_0 \\ x_1y_0 & y_1y_0 & z_1y_0 \\ x_1z_0 & y_1z_0 & z_1z_0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} \rightarrow \boxed{p^0 = R_1^0 p^1}$$





Example 3

- The vector v with coordinates $v^0 = (0, 1, 1)$ is rotated about y_0 by $\frac{\pi}{2}$.
- Find the coordinate of the resulting vector v_1 .

$$v_1^0 = R_{y, \frac{\pi}{2}} v^0$$
$$v_1^0 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$



Similarity Transformations

- A **similarity transformation** is used to convert the matrix representation of a linear transformation from one frame to another.
 - If A is the matrix representation of a linear transformation in frame $o_0x_0y_0z_0$ and
 - B is the representation of the same transformation in frame $o_1x_1y_1z_1$,
 - then A and B are related by:

$$B = (R_1^0)^{-1}AR_1^0$$

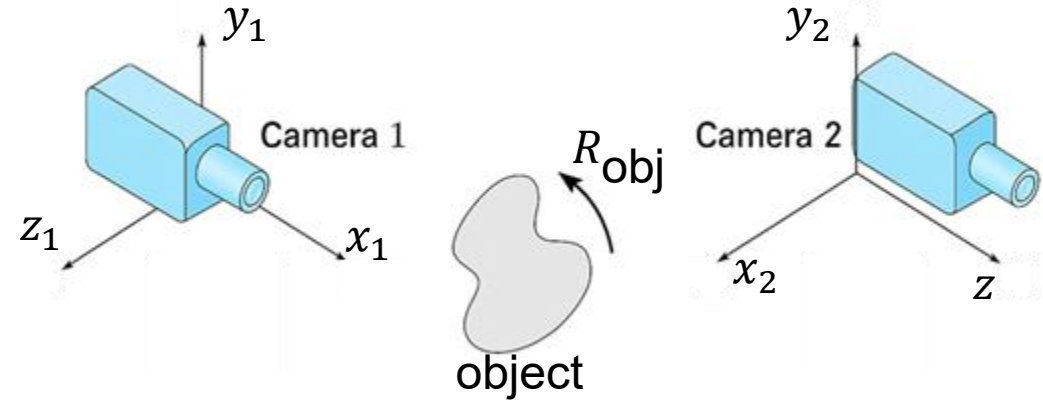


Example: Two Cameras

- **Camera 1** measures a rotation R_{obj}^1 .
- R_{obj}^2 , rotation as seen by camera 2.
- **Similarity transformation:**

$$R_{\text{obj}}^2 = (R_2^1)^{-1} R_{\text{obj}}^1 R_2^1$$

- The transformation changes only the reference frame, not the underlying rotation.
- Only the rotations are considered here; the relative positions of the cameras are neglected.





Example 4

- Suppose frames $o_0x_0y_0z_0$ and $o_1x_1y_1z_1$ are related by the rotation

$$R_1^0 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

- $A = R_{z,\theta}$, relative to the frame $o_0x_0y_0z_0$, then, relative to frame $o_1x_1y_1z_1$ we have

$$B = (R_1^0)^{-1}AR_1^0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

- **Note that:** B is a rotation of angle $-\theta$ around x_1 .



COMPOSITION OF ROTATIONS

- Recall that the matrix R_1^0 represents a rotational transformation between the frames $o_0x_0y_0z_0$ and $o_1x_1y_1z_1$.
- Suppose we now add a third coordinate frame $o_2x_2y_2z_2$ related to the frames $o_0x_0y_0z_0$ and $o_1x_1y_1z_1$ by rotational transformations.
- A given point p can then be represented by coordinates specified with respect to any of these three frames: p^0 , p^1 and p^2 .

- The relationship among these representations of p is

$$\begin{aligned} p^0 &= R_1^0 p^1, & p^1 &= R_2^1 p^2 \\ p^0 &= R_2^0 p^2 \end{aligned}$$

- Thus

$$p^0 = R_1^0 R_2^1 p^2 \rightarrow R_2^0 = R_1^0 R_2^1$$



COMPOSITION OF ROTATIONS

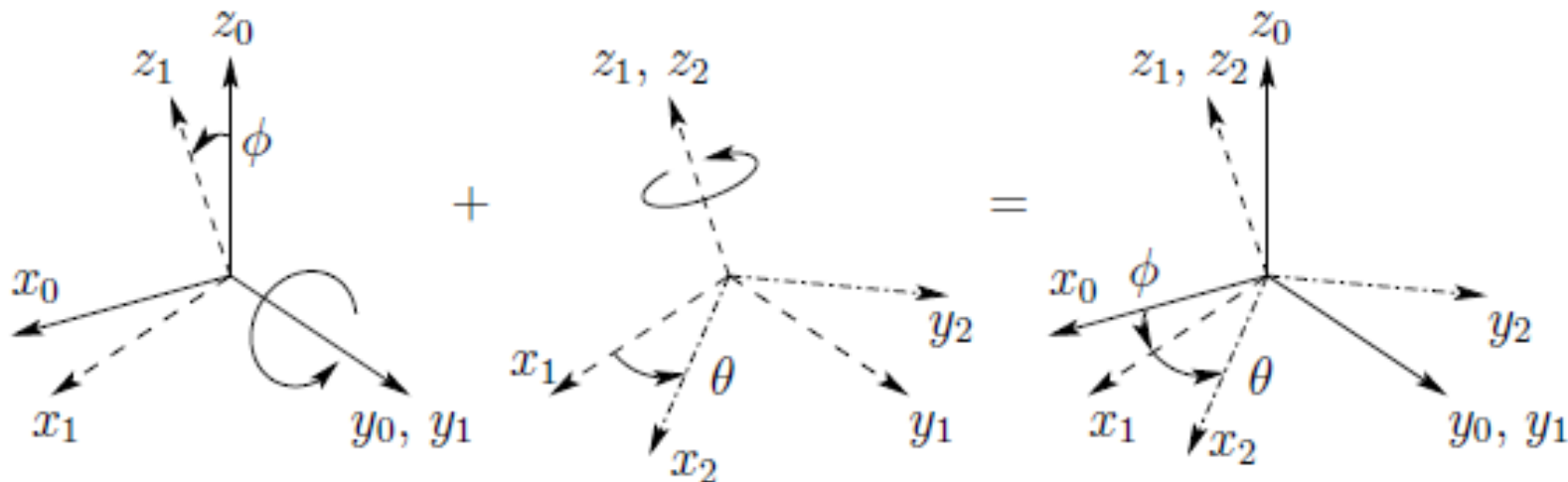
- Suppose initially that all three of the coordinate frames coincide.
- We first rotate the frame $o_2x_2y_2z_2$ relative to $o_0x_0y_0z_0$ according to the transformation R_1^0 .
- Then, with the frames $o_1x_1y_1z_1$ and frame $o_2x_2y_2z_2$ coincident, we rotate frame $o_2x_2y_2z_2$ relative to $o_1x_1y_1z_1$ according to the transformation R_2^1 .
- In each case we call the frame relative to which the rotation occurs the **current frame**.



Example 5

- A matrix R represents a rotation of angle ϕ about the current y-axis followed by a rotation of angle θ about the current z-axis.
- Find matrix R .

$$R = R_{y,\phi} R_{z,\theta} = \begin{bmatrix} c\phi & 0 & s\phi \\ 0 & 1 & 0 \\ -s\phi & 0 & c\phi \end{bmatrix} \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c\phi c\theta & -c\phi s\theta & s\phi \\ s\theta & c\theta & 0 \\ -s\phi c\theta & s\phi s\theta & c\phi \end{bmatrix}$$





Example 6

- Suppose that the above rotations are performed in the reverse order, that is, first a rotation about the current z-axis followed by a rotation about the current y-axis.
- Find rotation matrix R' .

$$R' = R_{z,\theta}R_{y,\phi} = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\phi & 0 & s\phi \\ 0 & 1 & 0 \\ -s\phi & 0 & c\phi \end{bmatrix} = \begin{bmatrix} c\theta c\phi & -s\theta & c\theta s\phi \\ s\theta c\phi & c\theta & s\theta s\phi \\ -s\phi & 0 & c\phi \end{bmatrix}$$

Remember that

$$R = R_{y,\phi}R_{z,\theta} = \begin{bmatrix} c\phi & 0 & s\phi \\ 0 & 1 & 0 \\ -s\phi & 0 & c\phi \end{bmatrix} \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c\phi c\theta & -c\phi s\theta & s\phi \\ s\theta & c\theta & 0 \\ -s\phi c\theta & s\phi s\theta & c\phi \end{bmatrix}$$

Thus

$$R \neq R'$$

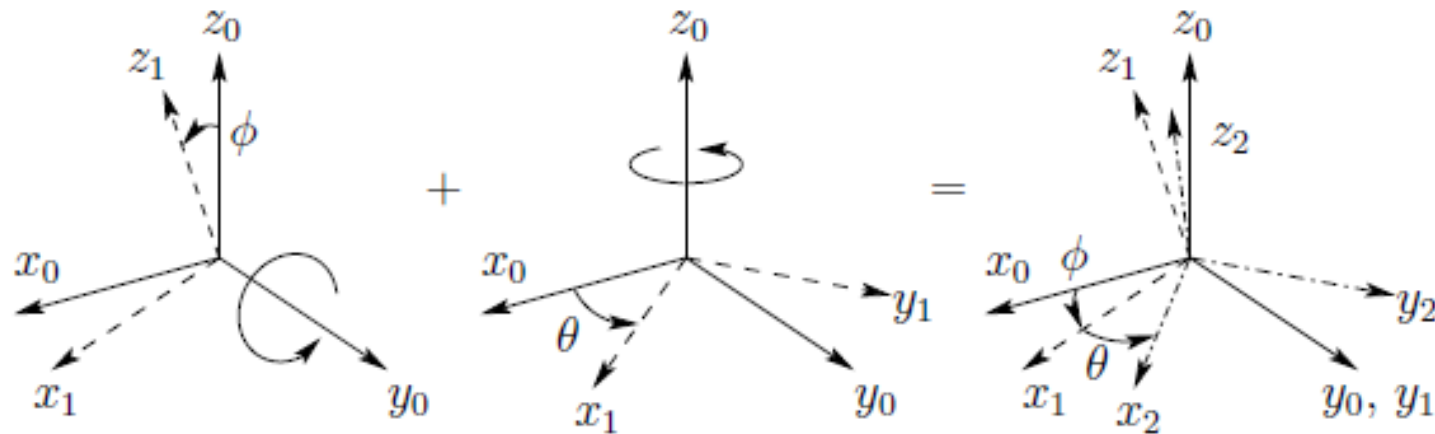
**Rotations do not
commutate!**



Rotation with respect to the fixed frame

- Many times, it is desired to perform a sequence of rotations, each about a given fixed coordinate frame, rather than about successive current frames.
- We refer to $o_0x_0y_0z_0$ as the fixed frame.
- The composition law given

$$R_2^0 = R_1^0[(R_1^0)^{-1}RR_1^0] = RR_1^0$$





Example 7

- Suppose that a rotation matrix R represents a rotation of angle ϕ about y_0 followed by a rotation of angle θ about the fixed z_0 .

$$R = R_{z,\theta} R_{y,\phi}$$
$$R = \begin{bmatrix} \cos \phi \cos \theta & -\sin \theta & \sin \phi \cos \theta \\ \cos \phi \sin \theta & \cos \theta & \sin \phi \sin \theta \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}$$



Example 8

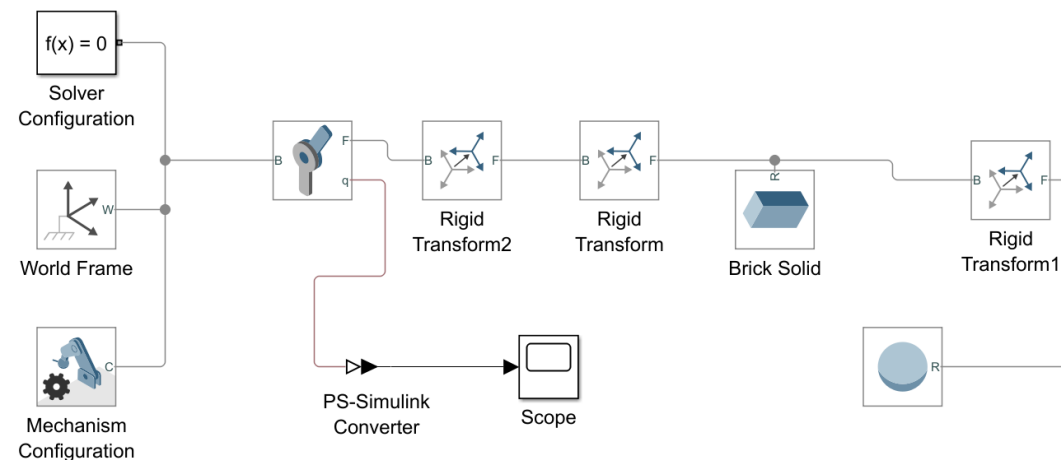
- Suppose R is defined by the following sequence of basic rotations in the order specified:
 1. A rotation of θ about the current x-axis
 2. A rotation of ϕ about the current z-axis
 3. A rotation of α about the fixed z-axis
 4. A rotation of β about the current y-axis
 5. A rotation of δ about the fixed x-axis

$$R = R_{x,\delta}R_{z,\alpha}R_{x,\theta}R_{z,\phi}R_{y,\beta}$$

Project: Model Using Simscape Multibody



- Model A Simple pendulum
 - Type smnew (create new Simscape Multibody model)
 - modify dimensions of the brick solid block [5 100 0] cm
 - Add revolute joint
 - Add rigid transform (rotate 5 deg around)
 - Add rigid transform (translate [0 -100/2 0])



Project: Model Using Simscape Multibody



- Model A Simple pendulum
 - Add rigid transform (translate [0 -100/2 0])
 - Add rigid transform (translate [0 -100/2 0])
 - Add sphere block (radius 5 cm)
 - Measure position of the revolute joint
 - Calculate the period of the pendulum osculation ($T = 2\pi\sqrt{L/g}$)

Project: Model Using Simscape Multibody



- Tutorial videos
 - [Physical Modeling Tutorial, Part 1: Introduction to Simscape \(youtube.com\)](#)



Attendance





Any Questions